

AI Virtual Assistant for Data Analysis & Prediction

Preface

Data analysis and predictive modeling are crucial for decision-making. This project integrates AI and machine learning to automate exploratory data analysis, ML predictions, and AI-generated insights using FastAPI, Streamlit, and PyTorch.

Introduction

Traditional data analysis requires manual effort. This project automates data processing, exploration, and prediction using AI-powered tools to make analytics accessible to all users.

Functionalities & Features

- File Upload & Data Handling (CSV/XLSX)
- Automated Exploratory Data Analysis (EDA)
- Machine Learning Predictions using PyTorch
- AI-Powered Insights with GPT-4
- Interactive AI Chatbot for Q&A

Industry Relevance & Applications

This project is applicable in data science, financial risk analysis, healthcare, business intelligence, and marketing for automated analytics and AI-driven decision-making.

Technology Stack

- Backend API: FastAPI (Asynchronous & High-Performance)
- Data Processing: Pandas, NumPy, PyTorch
- EDA & Visualization: Pandas Profiling, Seaborn, Plotly
- Machine Learning: PyTorch
- AI Insights: GPT-4 API, Hugging Face LLM
- Frontend UI: Streamlit
- Deployment: Docker + AWS/GCP

Conclusion

This AI-powered assistant simplifies data analytics by automating EDA, ML predictions, and AI

insights. It provides a comprehensive solution for industries requiring rapid, data-driven decision-making.